Statistical analyses

Paulet-Prats, A., Barceló-Coblijn, L., Cremades, E.

Abstract

The following presents the data from the different statistical analyses conducted in the study on functional categories and Williams syndrome.

Contents

1 Participants

TD participants				WS participants			
Group	Participant	Age	Sex	Group	Participant	Age	Sex
TD Catalan	DTCAT1	29	Female	WS Catalan	SWCAT1	12	Male
TD Catalan	DTCAT2	48	Female	WS Catalan	SWCAT2	15	Male
TD Catalan	DTCAT3	61	Female	WS Catalan	SWCAT3	9	Male
TD Catalan	DTCAT4	62	Female	WS Catalan	SWCAT4	9	Male
TD Catalan	DTCAT5	26	Female	WS Catalan	SWCAT5	17	Female
TD Spanish	DTESP1	52	Male	WS Spanish	SWESP1	23	Female
TD Spanish	DTESP2	32	Male	WS Spanish	SWESP2	57	Female
TD Spanish	DTESP3	13	Male	WS Spanish	SWESP3	36	Female
TD Spanish	DTESP4	16	Female	WS Spanish	SWESP4	16	Male
TD Spanish	DTESP5	22	Male	WS Spanish	SWESP5	44	Female

Table 1: Participant characteristics by group (TD on the left, WS on the right).

Variable	*t*	*df*	*p*	Hedges *g*	*SE g*
Age	1.59	18	.130	0.68	0.45

Table 2: Independent samples t-test for age.

2 MLUm

2.1 Grouping variable: Language

Table 3: Independent Samples T-Test

	t	df	p	Hedges' g	SE Hedges' g	95% CI for Hedges' g	
						Lower	Upper
MLUm	0.139	13.366	0.892	0.059	0.428	−0.818	0.935

2.2 Grouping variable: Williams Syndrome

Table 4: Independent Samples T-Test

	t	df	p	Hedges' g	SE Hedges' g	95% CI for Hedges' g	
						Lower	Upper
MLUm	3.389	18	0.003	1.451	0.529	0.441	2.431

3 Descriptive statistics and Shapiro–Wilk test

Table 5: Descriptive Statistics

	adv.		art. def.		art. ind.	
	0	1	0	1	0	1
Valid	10	10	10	10	10	10
Missing	0	0	0	0	0	0
Mode	57.000	22.000	47.000	19.000	19.000	2.000
Median	77.000	49.000	42.500	19.500	19.000	7.500
Mean	83.200	57.000	43.800	18.900	18.900	11.100
95% CI Mean Upper	96.370	76.784	51.948	23.828	23.615	17.176
95% CI Mean Lower	70.030	37.216	35.652	13.972	14.185	5.024
Std. Deviation	21.249	31.920	13.147	7.951	7.608	9.803
Shapiro-Wilk	0.927	0.912	0.907	0.965	0.904	0.848
P-value of Shapiro-Wilk	0.422	0.296	0.261	0.845	0.244	0.054
Minimum	57.000	22.000	28.000	4.000	8.000	2.000
Maximum	125.000	122.000	63.000	33.000	33.000	34.000

Table 6: Descriptive Statistics

	conj.		dem.	
	0	1	0	1
Valid	10	10	10	10
Missing	0	0	0	0
Mode	54.000	0.000	8.000	3.000
Median	71.000	38.000	7.500	2.000
Mean	71.100	37.000	7.000	4.100
95% CI Mean Upper	82.123	50.316	9.564	8.923
95% CI Mean Lower	60.077	23.684	4.436	-0.723
Std. Deviation	17.785	21.484	4.137	7.781
Shapiro-Wilk	0.855	0.986	0.967	0.504
P-value of Shapiro-Wilk	0.067	0.988	0.865	4.091×10^{-6}
Minimum	54.000	0.000	1.000	0.000
Maximum	103.000	69.000	14.000	26.000

Table 7: Descriptive Statistics

	interj.		num.		poss.		prep.	
	0	1	0	1	0	1	0	1
Valid	10	10	10	10	10	10	10	10
Missing	0	0	0	0	0	0	0	0
Mode	10.000	2.000	13.000	0.000	3.000	2.000	49.000	36.000
Median	8.000	7.000	12.500	4.000	3.500	2.500	74.500	39.000
Mean	7.300	9.100	10.900	4.400	5.000	4.900	74.300	38.600
95% CI Mean Upper	9.710	14.371	14.658	6.531	7.613	8.012	86.222	49.352
95% CI Mean Lower	4.890	3.829	7.142	2.269	2.387	1.788	62.378	27.848
Std. Deviation	3.889	8.504	6.064	3.438	4.216	5.021	19.236	17.347
Shapiro-Wilk	0.956	0.863	0.881	0.936	0.894	0.656	0.940	0.967
P-value of Shapiro-Wilk	0.735	0.082	0.133	0.512	0.188	2.626×10^{-4}	0.558	0.857
Minimum	0.000	1.000	3.000	0.000	0.000	2.000	49.000	4.000
Maximum	13.000	28.000	24.000	10.000	12.000	18.000	103.000	63.000

Table 8: Descriptive Statistics

	pron. fe.		pron. fo.		pron. rel.	
	0	1	0	1	0	1
Valid	10	10	10	10	10	10
Missing	0	0	0	0	0	0
Mode	14.000	18.000	3.000	5.000	7.000	0.000
Median	38.000	19.500	9.500	5.500	10.000	2.000
Mean	35.000	24.300	9.300	8.500	12.100	3.800
95% CI Mean Upper	41.902	33.795	13.220	14.338	15.835	6.719
95% CI Mean Lower	28.098	14.805	5.380	2.662	8.365	0.881
Std. Deviation	11.136	15.319	6.325	9.419	6.027	4.709
Shapiro-Wilk	0.930	0.892	0.905	0.801	0.872	0.796
P-value of Shapiro-Wilk	0.443	0.180	0.248	0.015	0.106	0.013
Minimum	14.000	1.000	2.000	0.000	4.000	0.000
Maximum	48.000	50.000	20.000	31.000	20.000	13.000

Table 9: Descriptive Statistics

	pron. int.		pron. ind.	
	0	1	0	1
Valid	10	10	10	10
Missing	0	0	0	0
Mode	0.000	1.000	5.000	2.000
Median	3.000	1.500	4.500	2.000
Mean	3.200	2.900	4.600	2.900
95% CI Mean Upper	4.996	6.026	6.124	4.284
95% CI Mean Lower	1.404	-0.226	3.076	1.516
Std. Deviation	2.898	5.043	2.459	2.234
Shapiro-Wilk	0.909	0.540	0.941	0.870
P-value of Shapiro-Wilk	0.275	1.073×10^{-5}	0.565	0.101
Minimum	0.000	0.000	1.000	0.000
Maximum	8.000	17.000	9.000	8.000

3.1 Levene's T-test

Table 10: Test of Equality of Variances (Levene's)

	F	df_1	df_2	p
adv.	1.427	1	18	0.248
art. def.	5.119	1	18	0.036
art. ind.	0.741	1	18	0.401
conj.	0.458	1	18	0.507
dem.	0.310	1	18	0.585
interj.	3.960	1	18	0.062
num.	1.722	1	18	0.206
poss.	0.008	1	18	0.930
prep.	0.237	1	18	0.632
pron. fe.	0.843	1	18	0.371
pron. fo.	0.208	1	18	0.654
pron. rel.	2.192	1	18	0.156
pron. int.	0.105	1	18	0.750
Pron. ind.	0.034	1	18	0.857

3.2 Student's T-tests

Table 11: Independent Samples T-Test

	t	df	p	Hedges' g	SE Hedges' g	95% CI for Hedges' g Lower	95% CI for Hedges' g Upper
adv.	2.161	18	0.044	0.925	0.472	-0.013	1.840
conj.	3.866	18	0.001	1.656	0.556	0.611	2.667
num.	2.949	18	0.009	1.263	0.507	0.281	2.216
prep.	4.358	18	3.788×10^{-4}	1.867	0.586	0.784	2.914
cli. pron.	1.787	18	0.091	0.765	0.459	-0.156	1.666

3.3 Welch's T-test

Table 12: Independent Samples T-Test

	t	df	p	Hedges' g	SE Hedges' g	95% CI for Hedges' g Lower	95% CI for Hedges' g Upper
art. def.	5.125	14.806	1.296×10^{-4}	2.195	0.636	0.999	3.350

3.4 Mann-Whitney T-test

Table 13: Independent Samples T-Test

	W	df	p	Rank-Biserial Correlation	SE	Lower	Upper
ind. art.	78.000	–	0.036	0.560	0.259	0.114	0.818
dem.	78.500	–	0.033	0.570	0.259	0.128	0.823
poss.	53.500	–	0.818	0.070	0.259	-0.421	0.529
pers. pron.	58.500	–	0.545	0.170	0.259	-0.334	0.598
rel. pron.	87.500	–	0.005	0.750	0.259	0.426	0.904
int. pron.	61.000	–	0.419	0.220	0.259	-0.287	0.630
ind. pron.	73.000	–	0.083	0.460	0.259	-0.021	0.768

4 Two-way factorial ANOVA

4.1 Adverb

Table 14: ANOVA - adv.

Cases	Sum of Squares	df	Mean Square	F	p	η^2	η^2
Language	561.800	1	561.800	0.905	0.356	0.034	0.054
SW	3432.200	1	3432.200	5.528	0.032	0.206	0.257
Language * SW	2737.800	1	2737.800	4.410	0.052	0.164	0.216
Residuals	9934.000	16	620.875				

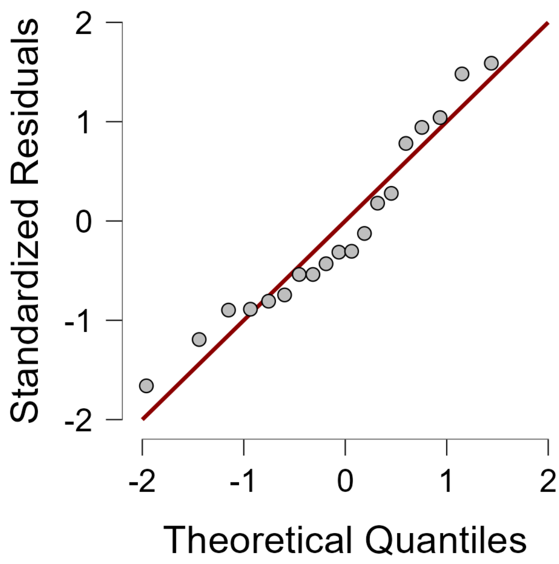

Figure 1: Adverb Q-Q

Table 15: Post Hoc Comparisons - Language

		Mean Difference	SE	t	p_{bonf}
0	1	10.600	11.143	0.951	0.356

Table 16: Post Hoc Comparisons - Language * SW

		Mean Difference	SE	t	p_{bonf}
0 0	1 0	−12.800	15.759	−0.812	1.000
	0 1	2.800	15.759	0.178	1.000
	1 1	36.800	15.759	2.335	0.197
1 0	0 1	15.600	15.759	0.990	1.000
	1 1	49.600	15.759	3.147	0.037
0 1	1 1	34.000	15.759	2.157	0.279

Table 17: Post Hoc Comparisons - SW

		Mean Difference	SE	t	p_{bonf}
0	1	26.200	11.143	2.351	0.032

4.2 Definite Article

Table 18: ANOVA - Def. Art.

Cases	Sum of Squares	df	Mean Square	F	p	η^2	η^2
SW	3100.050	1	3100.050	24.875	1.342×10^{-4}	0.593	0.609
Language	54.450	1	54.450	0.437	0.518	0.010	0.027
SW * Language	76.050	1	76.050	0.610	0.446	0.015	0.037
Residuals	1994.000	16	124.625				

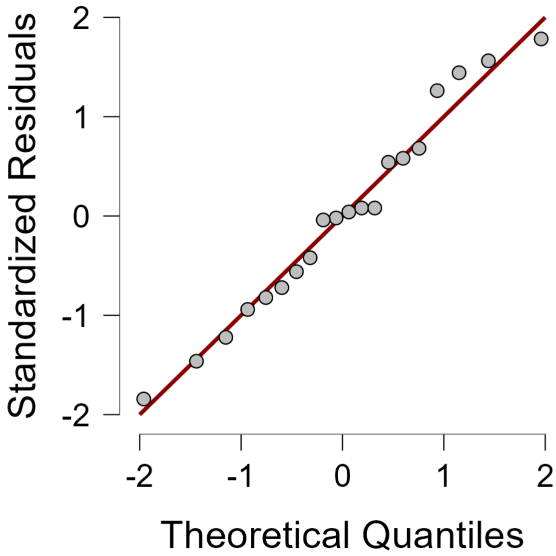

Figure 2: Definite Article Q-Q

Table 19: Post Hoc Comparisons - SW * Language

		Mean Difference	SE	t	p_{bonf}
0 0	1 0	28.800	7.060	4.079	0.005
	0 1	7.200	7.060	1.020	1.000
	1 1	28.200	7.060	3.994	0.006
1 0	0 1	−21.600	7.060	−3.059	0.045
	1 1	−0.600	7.060	−0.085	1.000
0 1	1 1	21.000	7.060	2.974	0.054

Table 20: Post Hoc Comparisons - SW

		Mean Difference	SE	t	p_{bonf}
0	1	24.900	4.992	4.987	1.342×10^{-4}

Table 21: Post Hoc Comparisons - Language

		Mean Difference	SE	t	p_{bonf}
0	1	3.300	4.992	0.661	0.518

4.3 Indefinite Article

Table 22: ANOVA - Ind. Art.

Cases	Sum of Squares	df	Mean Square	F	p	η^2	η^2
SW	304.200	1	304.200	3.788	0.069	0.180	0.191
Language	88.200	1	88.200	1.098	0.310	0.052	0.064
SW * Language	12.800	1	12.800	0.159	0.695	0.008	0.010
Residuals	1284.800	16	80.300				

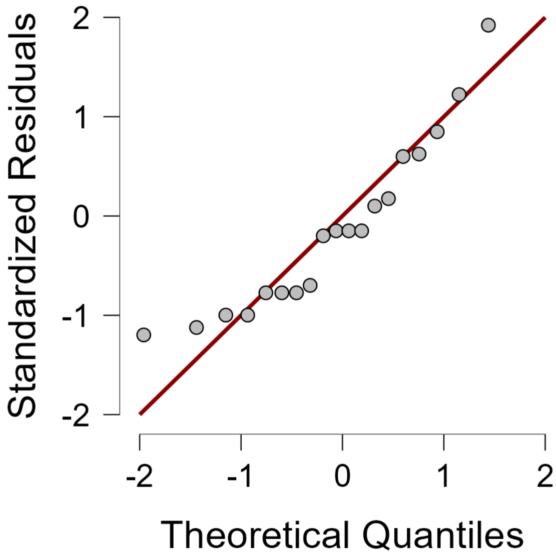

Figure 3: Indefinite Article Q-Q

Table 23: Post Hoc Comparisons - SW

		Mean Difference	SE	t	p_{bonf}
0	1	7.800	4.007	1.946	0.069

Table 24: Post Hoc Comparisons - Language

		Mean Difference	SE	t	p_{bonf}
0	1	4.200	4.007	1.048	0.310

Table 25: Post Hoc Comparisons - SW * Language

		Mean Difference	SE	t	p_{bonf}
0 0	1 0	6.200	5.667	1.094	1.000
	0 1	2.600	5.667	0.459	1.000
	1 1	12.000	5.667	2.117	0.301
1 0	0 1	−3.600	5.667	−0.635	1.000
	1 1	5.800	5.667	1.023	1.000
0 1	1 1	9.400	5.667	1.659	0.700

4.4 Conjunctions

Table 26: ANOVA - conj.

Cases	Sum of Squares	df	Mean Square	F	p	η^2	η^2
SW	5814.050	1	5814.050	13.910	0.002	0.454	0.465
Language	61.250	1	61.250	0.147	0.707	0.005	0.009
SW * Language	252.050	1	252.050	0.603	0.449	0.020	0.036
Residuals	6687.600	16	417.975				

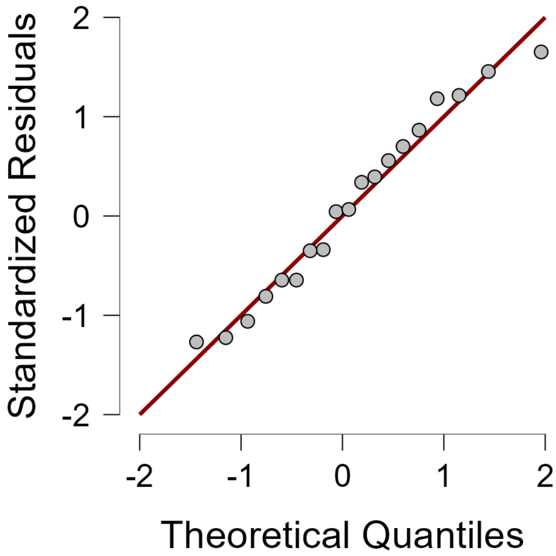

Figure 4: Conjunctions Q-Q

Table 27: Post Hoc Comparisons - SW

		Mean Difference	SE	t	p_{bonf}
0	1	34.100	9.143	3.730	0.002

Table 28: Post Hoc Comparisons - Language

		Mean Difference	SE	t	p_{bonf}
0	1	−3.500	9.143	−0.383	0.707

Table 29: Post Hoc Comparisons - SW * Language

		Mean Difference	SE	t	p_{bonf}
0 0	1 0	27.000	12.930	2.088	0.319
	0 1	−10.600	12.930	−0.820	1.000
	1 1	30.600	12.930	2.367	0.185
1 0	0 1	−37.600	12.930	−2.908	0.062
	1 1	3.600	12.930	0.278	1.000
0 1	1 1	41.200	12.930	3.186	0.034

4.5 Demonstratives

Table 30: ANOVA - dem.

Cases	Sum of Squares	df	Mean Square	F	p	η^2	η^2
SW	42.050	1	42.050	1.044	0.322	0.057	0.061
Language	18.050	1	18.050	0.448	0.513	0.024	0.027
SW * Language	36.450	1	36.450	0.905	0.356	0.049	0.054
Residuals	644.400	16	40.275				

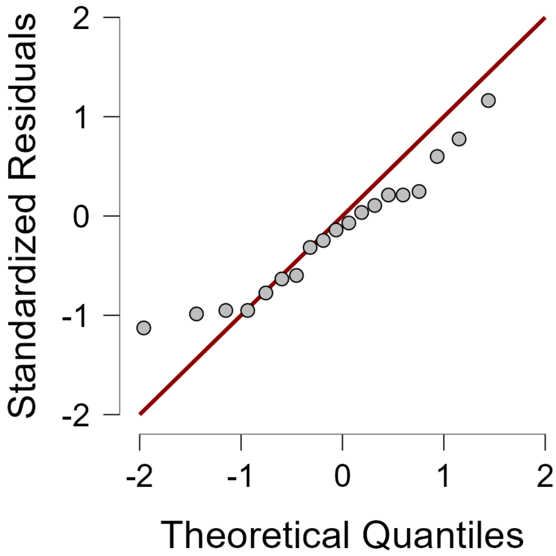

Figure 5: Demonstratives Q-Q

Table 31: Post Hoc Comparisons - SW

		Mean Difference	SE	t	p_{bonf}
0	1	2.900	2.838	1.022	0.322

Table 32: Post Hoc Comparisons - Language

		Mean Difference	SE	t	p_{bonf}
0	1	1.900	2.838	0.669	0.513

4.6 Interjections

Table 33: ANOVA - interj.

Cases	Sum of Squares	df	Mean Square	F	p	η^2	η^2
SW	16.200	1	16.200	0.368	0.553	0.020	0.022
Language	1.800	1	1.800	0.041	0.842	0.002	0.003
SW * Language	80.000	1	80.000	1.815	0.197	0.100	0.102
Residuals	705.200	16	44.075				

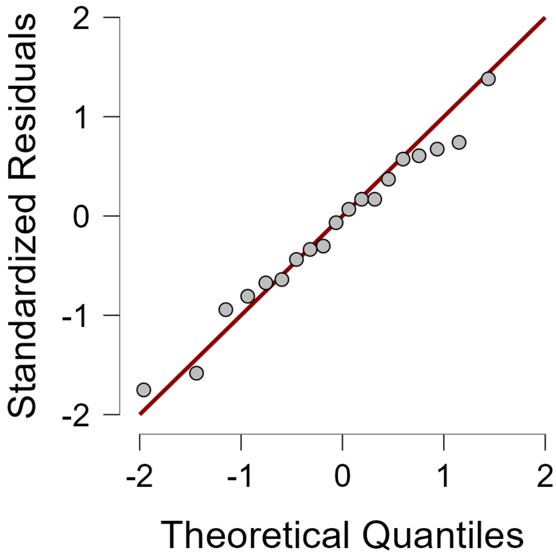

Figure 6: Interjections Q-Q

Table 34: Post Hoc Comparisons - SW

		Mean Difference	SE	t	p_{bonf}
0	1	−1.800	2.969	−0.606	0.553

Table 35: Post Hoc Comparisons - Language

		Mean Difference	SE	t	p_{bonf}
0	1	0.600	2.969	0.202	0.842

Table 36: Post Hoc Comparisons - SW * Language

		Mean Difference	SE	t	p_{bonf}
0 0	1 0	−5.800	4.199	−1.381	1.000
	0 1	−3.400	4.199	−0.810	1.000
	1 1	−1.200	4.199	−0.286	1.000
1 0	0 1	2.400	4.199	0.572	1.000
	1 1	4.600	4.199	1.096	1.000
0 1	1 1	2.200	4.199	0.524	1.000

4.7 Numerals

Table 37: ANOVA - num.

Cases	Sum of Squares	df	Mean Square	F	p	η^2	η^2
SW	211.250	1	211.250	9.918	0.006	0.326	0.383
Language	54.450	1	54.450	2.556	0.129	0.084	0.138
SW * Language	42.050	1	42.050	1.974	0.179	0.065	0.110
Residuals	340.800	16	21.300				

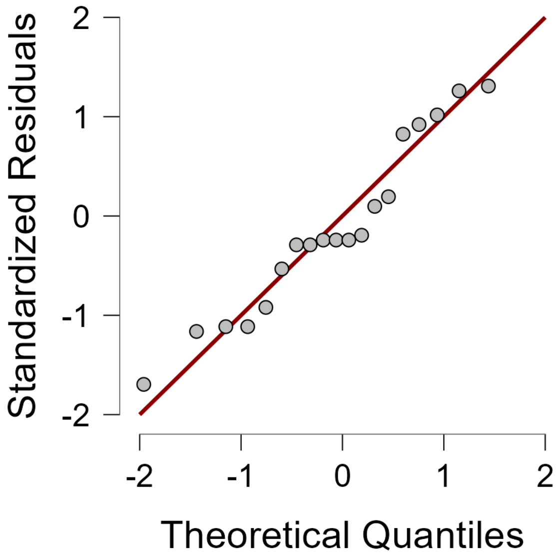

Figure 7: Numerals Q-Q

Table 38: Post Hoc Comparisons - SW

		Mean Difference	SE	t	p_{bonf}
0	1	6.500	2.064	3.149	0.006

Table 39: Post Hoc Comparisons - Language

		Mean Difference	SE	t	p_{bonf}
0	1	3.300	2.064	1.599	0.129

Table 40: Post Hoc Comparisons - SW * Language

		Mean Difference	SE	t	p_{bonf}
0 0	1 0	9.400	2.919	3.220	0.032
	0 1	6.200	2.919	2.124	0.298
	1 1	9.800	2.919	3.357	0.024
1 0	0 1	−3.200	2.919	−1.096	1.000
	1 1	0.400	2.919	0.137	1.000
0 1	1 1	3.600	2.919	1.233	1.000

4.8 Possessives

Table 41: ANOVA - poss.

Cases	Sum of Squares	df	Mean Square	F	p	η^2	η^2
SW	0.050	1	0.050	0.002	0.963	1.292×10^{-4}	1.387×10^{-4}
Language	26.450	1	26.450	1.174	0.295	0.068	0.068
SW * Language	0.050	1	0.050	0.002	0.963	1.292×10^{-4}	1.387×10^{-4}
Residuals	360.400	16	22.525				

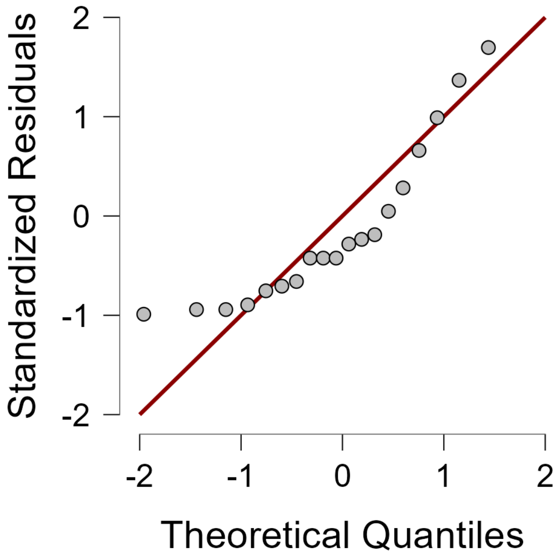

Figure 8: Possessives Q-Q

Table 42: Post Hoc Comparisons - SW

		Mean Difference	SE	t	p_{bonf}
0	1	0.100	2.122	0.047	0.963

Table 43: Post Hoc Comparisons - Language

		Mean Difference	SE	t	p_{bonf}
0	1	2.300	2.122	1.084	0.295

Table 44: Post Hoc Comparisons - SW * Language

		Mean Difference	SE	t	p_{bonf}
0 0	1 0	0.200	3.002	0.067	1.000
	0 1	2.400	3.002	0.800	1.000
	1 1	2.400	3.002	0.800	1.000
1 0	0 1	2.200	3.002	0.733	1.000
	1 1	2.200	3.002	0.733	1.000
0 1	1 1	1.388×10^{-17}	3.002	4.623×10^{-18}	1.000

4.9 Prepositions

Table 45: ANOVA - prep.

Cases	Sum of Squares	df	Mean Square	F	p	η^2	η^2
SW	6372.450	1	6372.450	18.260	5.823×10^{-4}	0.513	0.533
Language	110.450	1	110.450	0.316	0.582	0.009	0.019
SW * Language	344.450	1	344.450	0.987	0.335	0.028	0.058
Residuals	5583.600	16	348.975				

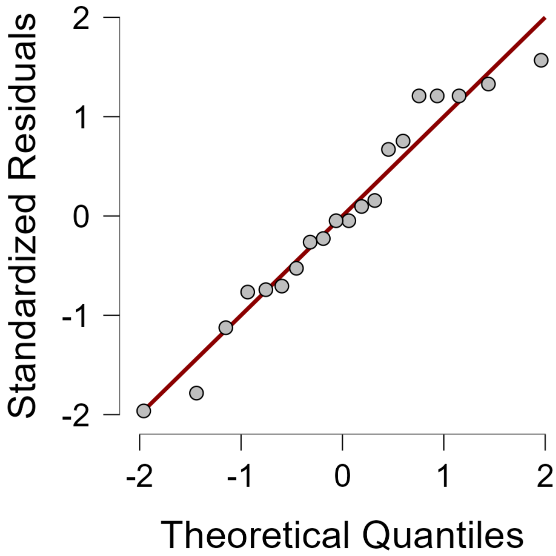

Figure 9: Prepositions Q-Q

Table 46: Post Hoc Comparisons - SW

		Mean Difference	SE	t	p_{bonf}
0	1	35.700	8.354	4.273	5.823×10^{-4}

Table 47: Post Hoc Comparisons - Language

		Mean Difference	SE	t	p_{bonf}
0	1	4.700	8.354	0.563	0.582

Table 48: Post Hoc Comparisons - SW * Language

		Mean Difference	SE	t	p_{bonf}
0 0	1 0	44.000	11.815	3.724	0.011
	0 1	13.000	11.815	1.100	1.000
	1 1	40.400	11.815	3.419	0.021
1 0	0 1	−31.000	11.815	−2.624	0.111
	1 1	−3.600	11.815	−0.305	1.000
0 1	1 1	27.400	11.815	2.319	0.204

4.10 Personal Pronouns

Table 49: ANOVA - Pers. Pron.

Cases	Sum of Squares	df	Mean Square	F	p	η^2	η^2
SW	3.200	1	3.200	0.050	0.826	0.003	0.003
Language	12.800	1	12.800	0.201	0.660	0.011	0.012
SW * Language	125.000	1	125.000	1.959	0.181	0.108	0.109
Residuals	1020.800	16	63.800				

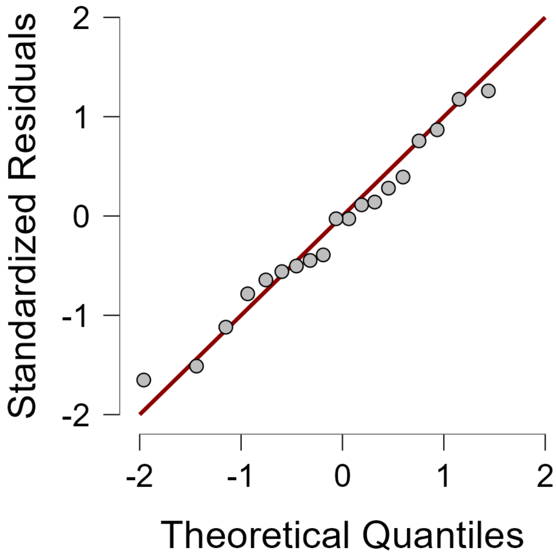

Figure 10: Personal Pronouns Q-Q

Table 50: Post Hoc Comparisons - SW

		Mean Difference	SE	t	p_{bonf}
0	1	0.800	3.572	0.224	0.826

Table 51: Post Hoc Comparisons - Language

		Mean Difference	SE	t	p_{bonf}
0	1	1.600	3.572	0.448	0.660

Table 52: Post Hoc Comparisons - SW * Language

		Mean Difference	SE	t	p_{bonf}
0 0	1 0	−4.200	5.052	−0.831	1.000
	0 1	−3.400	5.052	−0.673	1.000
	1 1	2.400	5.052	0.475	1.000
1 0	0 1	0.800	5.052	0.158	1.000
	1 1	6.600	5.052	1.306	1.000
0 1	1 1	5.800	5.052	1.148	1.000

4.11 Relative Pronouns

Table 53: ANOVA - rel. pron.

Cases	Sum of Squares	df	Mean Square	F	p	η^2	η^2
SW	344.450	1	344.450	12.161	0.003	0.395	0.432
Language	31.250	1	31.250	1.103	0.309	0.036	0.065
SW * Language	42.050	1	42.050	1.485	0.241	0.048	0.085
Residuals	453.200	16	28.325				

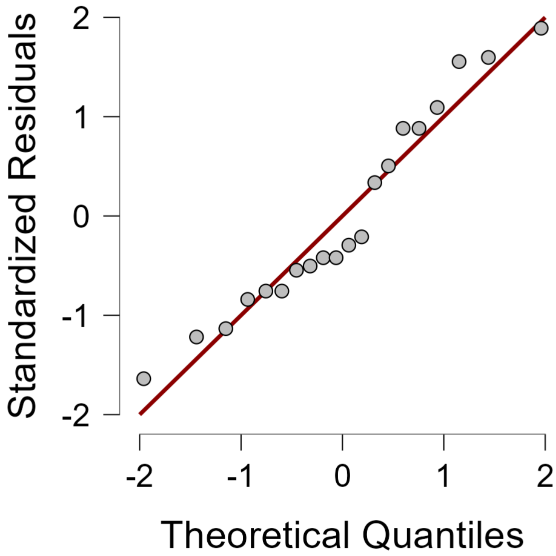

Figure 11: Relative Pronouns Q-Q

Table 54: Post Hoc Comparisons - SW

		Mean Difference	SE	t	p_{bonf}
0	1	8.300	2.380	3.487	0.003

Table 55: Post Hoc Comparisons - Language

		Mean Difference	SE	t	p_{bonf}
0	1	−2.500	2.380	−1.050	0.309

Table 56: Post Hoc Comparisons - SW * Language

		Mean Difference	SE	t	p_{bonf}
0 0	1 0	5.400	3.366	1.604	0.769
	0 1	−5.400	3.366	−1.604	0.769
	1 1	5.800	3.366	1.723	0.625
1 0	0 1	−10.800	3.366	−3.209	0.033
	1 1	0.400	3.366	0.119	1.000
0 1	1 1	11.200	3.366	3.327	0.026

4.12 Interrogatives

Table 57: ANOVA - Int. pron.

Cases	Sum of Squares	df	Mean Square	F	p	η^2	η^2
SW	0.450	1	0.450	0.027	0.872	0.001	0.002
Language	36.450	1	36.450	2.179	0.159	0.120	0.120
SW * Language	0.450	1	0.450	0.027	0.872	0.001	0.002
Residuals	267.600	16	16.725				

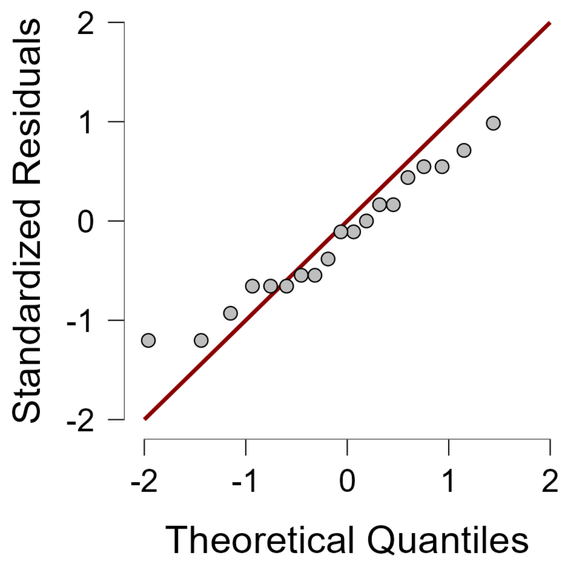

Figure 12: Interrogatives Q-Q

Table 58: Post Hoc Comparisons - SW

		Mean Difference	SE	t	p_{bonf}
0	1	0.300	1.829	0.164	0.872

Table 59: Post Hoc Comparisons - Language

		Mean Difference	SE	t	p_{bonf}
0	1	2.700	1.829	1.476	0.159

Table 60: Post Hoc Comparisons - SW * Language

		Mean Difference	SE	t	p_{bonf}
0 0	1 0	1.110×10^{-16}	2.587	4.292×10^{-17}	1.000
	0 1	2.400	2.587	0.928	1.000
	1 1	3.000	2.587	1.160	1.000
1 0	0 1	2.400	2.587	0.928	1.000
	1 1	3.000	2.587	1.160	1.000
0 1	1 1	0.600	2.587	0.232	1.000

4.13 Indefinite Pronouns

Table 61: ANOVA - Ind. Pron.

Cases	Sum of Squares	df	Mean Square	F	p	η^2	η^2
SW	14.450	1	14.450	2.558	0.129	0.127	0.138
Language	8.450	1	8.450	1.496	0.239	0.074	0.085
SW * Language	0.450	1	0.450	0.080	0.781	0.004	0.005
Residuals	90.400	16	5.650				

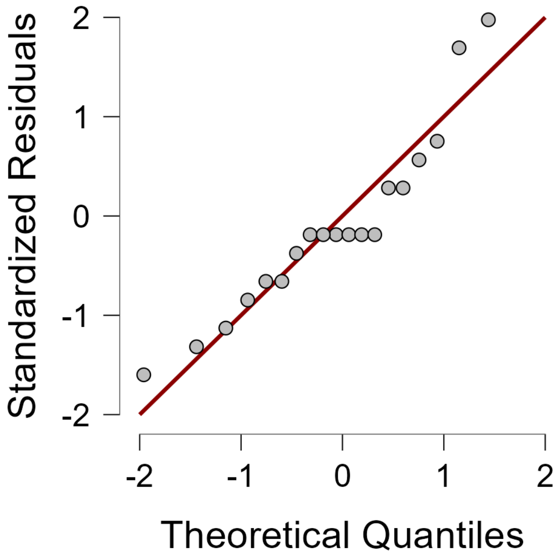

Figure 13: Indefinite Pronouns Q-Q

Table 62: Post Hoc Comparisons - SW

		Mean Difference	SE	t	p_{tukey}
0	1	1.700	1.063	1.599	0.129

Table 63: Post Hoc Comparisons - Language

		Mean Difference	SE	t	p_{tukey}
0	1	−1.300	1.063	−1.223	0.239

Table 64: Post Hoc Comparisons - SW * Language

		Mean Difference	SE	t	p_{tukey}
0 0	1 0	1.400	1.503	0.931	0.789
	0 1	−1.600	1.503	−1.064	0.715
	1 1	0.400	1.503	0.266	0.993
1 0	0 1	−3.000	1.503	−1.996	0.230
	1 1	−1.000	1.503	−0.665	0.909
0 1	1 1	2.000	1.503	1.330	0.558

5 Bootstraping Analysis

Table 65: Bootstrap Specifications

Parameter	Value
Sampling Method	Simple
Number of Samples	2000
Confidence Interval Level	95.0%
Confidence Interval Type	Percentile

Table 66: Between-Subjects Factors

Factor	N
SW = 0	10
SW = 1	10
Language = 0	10
Language = 1	10

Table 67: Multivariate Tests

Effect	Test	Value	F	Hypothesis df	Error df	p
Intercept	Pillai's Trace	0.991	34.287	13.000	4.000	0.002
Intercept	Wilks' Lambda	0.009	34.287	13.000	4.000	0.002
Intercept	Hotelling's Trace	111.434	34.287	13.000	4.000	0.002
Intercept	Roy's Largest Root	111.434	34.287	13.000	4.000	0.002
SW	Pillai's Trace	0.964	8.256	13.000	4.000	0.028
SW	Wilks' Lambda	0.036	8.256	13.000	4.000	0.028
SW	Hotelling's Trace	26.833	8.256	13.000	4.000	0.028
SW	Roy's Largest Root	26.833	8.256	13.000	4.000	0.028
Language	Pillai's Trace	0.555	0.383	13.000	4.000	0.915
Language	Wilks' Lambda	0.445	0.383	13.000	4.000	0.915
Language	Hotelling's Trace	1.246	0.383	13.000	4.000	0.915
Language	Roy's Largest Root	1.246	0.383	13.000	4.000	0.915
SW × Language	Pillai's Trace	0.552	0.378	13.000	4.000	0.918
SW × Language	Wilks' Lambda	0.448	0.378	13.000	4.000	0.918
SW × Language	Hotelling's Trace	1.230	0.378	13.000	4.000	0.918
SW × Language	Roy's Largest Root	1.230	0.378	13.000	4.000	0.918

Table 68: Tests of Between-Subjects Effects

Effect	DV	SS	df	MS	F	p
	adv.	6731.800	3	2243.933	3.614	.036
	art. def.	3230.550	3	1076.850	8.641	.001
	art. ind.	405.200	3	135.067	1.682	.211
	conj.	6127.350	3	2042.450	4.887	.013
	dem.	96.550	3	32.183	0.799	.512
	interj.	98.000	3	32.667	0.741	.543
Corrected Model	num.	307.750	3	102.583	4.816	.014
	poss.	26.550	3	8.850	0.393	.760
	prep.	6827.350	3	2275.783	6.521	.004
	pron. fo.	141.000	3	47.000	0.737	.545
	pron. rel.	417.750	3	139.250	4.916	.013
	pron. int.	37.350	3	12.450	0.744	.541
	pron. ind.	23.350	3	7.783	1.378	.286
	adv.	3432.200	1	3432.200	5.528	.032
	art. def.	3100.050	1	3100.050	24.875	¡.001
	art. ind.	304.200	1	304.200	3.788	.069
	conj.	5814.050	1	5814.050	13.910	.002
	dem.	42.050	1	42.050	1.044	.322
	interj.	16.200	1	16.200	0.368	.553
SW	num.	211.250	1	211.250	9.918	.006
	poss.	0.050	1	0.050	0.002	.963
	prep.	6372.450	1	6372.450	18.260	.001
	pron. fo.	3.200	1	3.200	0.050	.826
	pron. rel.	344.450	1	344.450	12.161	.003
	pron. int.	0.450	1	0.450	0.027	.872
	pron. ind.	14.450	1	14.450	2.558	.129
	adv.	561.800	1	561.800	0.905	.356
	art. def.	54.450	1	54.450	0.437	.518
	art. ind.	88.200	1	88.200	1.098	.310
	conj.	61.250	1	61.250	0.147	.707
	dem.	18.050	1	18.050	0.448	.513
	interj.	1.800	1	1.800	0.041	.842
Language	num.	54.450	1	54.450	2.556	.129
	poss.	26.450	1	26.450	1.174	.295
	prep.	110.450	1	110.450	0.316	.582
	pron. fo.	12.800	1	12.800	0.201	.660
	pron. rel.	31.250	1	31.250	1.103	.309
	pron. int.	36.450	1	36.450	2.179	.159
	pron. ind.	8.450	1	8.450	1.496	.239
	adv.	2737.800	1	2737.800	4.410	.052
	art. def.	76.050	1	76.050	0.610	.446
	art. ind.	12.800	1	12.800	0.159	.695
	conj.	252.050	1	252.050	0.603	.449
	dem.	36.050	1	36.050	0.905	.356
	interj.	80.000	1	80.000	1.815	.197
SW × Language	num.	42.050	1	42.050	1.974	.179
	poss.	0.050	1	0.050	0.002	.963
	prep.	344.450	1	344.450	0.987	.335
	pron. fo.	125.000	1	125.000	1.959	.181
	pron. rel.	42.050	1	42.050	1.485	.241
	pron. int.	0.450	1	0.450	0.027	.872
	pron. ind.	0.450	1	0.450	0.080	.781